

F24-220-R-NAZAR

Project Team

Tauheed Akbar	21I-0723
Taha Ahmed Pirzada	21I-0570
Nouman Ahmad	21I-0727

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Supervised by

Dr. Isma Ul Hassan



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Chapter 1

Introduction

The purpose of our project proposal is to provide the detail of the development of our FYP project NAZAR's mobile application designed to enhance eye health screening through advanced image processing technology. This application not only addresses the cataract but also an eye's refractive error assessment, by using the smart phone cameras, and its torch and AI models and machine learning. Cataracts and refractive errors, such as myopia, hypermetropia, and astigmatism, are the main cause of visual impairment and blindness globally. By adding these diagnostic capabilities into a single application, we can provide a cost-effective, readily available and accessible solution for early detection of the above mentioned eye condition's and refractive errors, particularly in under-served and rural areas in our country Pakistan.

1.1 Problem Domain

The progress made in the technology sector especially in the area of mobile technologies and AI models has led the widespread usage of these technologies worldwide and so much so they are part of our lives and livelihood. Using these widespread resources, medical field has come a long way, healthcare system has also increased its dependency on technologies day by day. Despite the wide adoption of technologies in medical sector, the equipment's related to these are quite expensive and ophthalmologist need special training to use it and civilians cannot afford it. Thus remains a significant gap in the area of cost effective, and scalable solution using everyday technologies. Thus is a barrier in our society for early detection of eye diseases specially cataract, and other eye refractive diseases. In this context, mobile applications leveraging AI models to detect and processed images taken from mobile phones camera. This approach provide cost effect, solution and less training required to use it thus easily adoptable in medical field and easily accessible for normal people to get early detection of eye diseases, as the application can give you the

result instantly about potential conditions.

1.2 Research Problem Statement

The RND based FYP project aims to develop a mobile application using the power of AI model in its back-end to preprocess the image and scan the image using the already trained model, the image taken for it is taken by smart phone camera. The specific problem that is required to be addressed is the increase in the reliability and accuracy of the application and remove the need for extra equipment needed to attach to the smartphone like fundus camera, that are not accessible for normal civilians, another issue is lack of accessible equipment in the areas like underserved rural areas, in which an easy portable device is required.

The main gap in the field is the adoption of better and accurate AI models for the mobile applications, thus the need is to tackle the variation in image quality due to difference in the image quality of each camera in different smartphones.

By addressing these above gaps in the field this project NAZAR of ours is to tackle and contribute to the development of a cost-effective solution.

1.3 Software Requirements Specification (if applicable)

In our project NAZAR we are developing a cross-platform (iOS and Android) Android application that utilizes an AI model to detect eye disease from images captured from a mobile camera, for this we have outlined functional and non-functional requirements below for our application.

1.3.1 Functional Requirements

Below are the functional requirements:

Image capturing and preprocessing:

Features: Capable of allowing user to capture image or upload images from smart phone cameras.

Constraints: Application must guide user to capture images of their eyes clearly.

Behaviour: The application should give feedback like eye positioning to ensure good quality images.

Performance: the system should be able to process images using inbuilt processors.

AI model:

Features: The AI model should be capable to analyze images and detect eye diseases like cataract.

Constraints: The AI model should work on the on device chipsets.

Behaviour: After analyzing the image, the model should provide a basic feedback about its predictions

Performance: The AI model should give an accuracy rate of its result.

Data Management:

Features: the application should be capable of storing images for further boosting the dataset.

Constraints: The data stored must comply with privacy laws of Pakistan.

Behaviour: The application should secure the user data and should encrypt it, when transmitting to server.

Performance: Should be capable of performing data retrieval to ensure storing of past history of patients.

Map Module:

Features: Display a map of all eye clinics to help guide user to nearest centre.

Constraints: Map must be responsive.

Behaviour: Map can be zoom in and out, and be able to pan on the map

Performance: Map module should be quickly loadable without any noticeable lags.

AR in Camera module:

Features: Use AR to guide users in positioning their eyes.

Constraints: AR must be supportable for all types of camera in smartphone, and both back and front camera as well.

Behaviour: If eyes are not aligned properly the app should notify the user.

Performance: Real-time feedback should be quick and instant, without any noticeable delays.

Profile Module:

Features: Allow user to create and manage profiles, and keep history of patients and user.

Constraints: There should be multiple optionality and variation of profile management for NGO's , Hospitals and normal users.

Behaviour: User should be able to load the history, and Clinics can keep the patients record, and NGOs can track their work and dataset.

Performance: Data syncing should be quick when online.

Offline functionality Module:

Features: Allow NGO's to save patients data offline and access application without active internet.

Constraints: Should allow syncing when device is online and proper optimization of

storage.

Behaviour: If app detects no internet connection it should notify user, and switch to offline functionality.

Performance: the eye scanning should not be hindered and the ai model should provide accuracy comparable to online.

Tool Tip Module:

Features: provide guidance via tool tips and tutorial for new users.

Constraints: Must not interfere or obstruct the module when explaining it.

Behaviour: Tutorial should pop up when a new user opens the application.

Performance: the transition between tool tips should be smooth and the functionality should load instantaneously.

1.3.2 Non-Functional Requirements

The non functional requirement outlines the performance, security, scalability and reliability criteria that software must meet :

Performance: The result of image scan should be given under few seconds, on mid to high end mobile devices using their fast chipsets for processing.

Security: The application must provide user with at least basic level of security like for example their on device application information should be encrypted and prevents unauthorized access or disclosure and use and gather information of user's with their consent.

Scalability: The AI model should be able to scaled gradually in order to incorporate better sensitivity and accuracy in its result. The application should be able to increase its user base.

Portability: The application should be compatible with wide range of smart phones across multiple platforms like (IOS AND ANDROID).

Chapter 2

Literature Review

2.1 Related Research

Below are several literature that were reviewed for the MID report.

2.1.1 Research Item 1

Askarian et al., 2021

2.1.1.1 Summary of the research item

In this research study Askarian and colleagues have proposed a portable, easy to use telemedicine solution for detecting cataracts with more than 95 percent accuracy using smartphone captured pictures, they have done this in order to find the feasibility of detecting cataracts using smartphones. To accomplish this they have developed a novel luminance based method for feature extraction in which instead of regular rgb, they use the brightness and luminance values of an image in order to extract features for cataract classification. Their methodology is to first capture an image of an eye using a smartphone camera from 10 cm to 50 cm distance, then the image goes through preprocessing including a median filter and watershed transformation to take out artifacts. Next luminance values are calculated based on the specific phones camera sensor characteristics, the converted image is then sent to a support vector machine algorithm which then classifies the presence of cataract within the eye. They trained the model on 100 pictures of artificial eye models, in which 50 were cataracts and 50 were non cataracts, and then used 10 fold cross validation for the dataset to reduce chances of overfitting. They found that this method not only detects cataracts with 96.6 percent accuracy but also that environmental factors such as lighting conditions, camera angle and distance, and what smartphone

is used has negligible impact on classification, making this method a very versatile and practical approach for real life detection.

2.1.1.2 Critical analysis of the research item

Strengths:

1. The model is trained on data from different smartphones and environmental conditions making it more versatile
2. This method uses a novel luminance based approach for feature extraction rather than rgb, providing higher accuracy than other rgb based approaches
3. The model achieved a very high 96.6 percent accuracy on a small dataset using only a support vector machine algorithm without a CNN, making the computation a lot less and the process more simple and easy

Weaknesses:

1. The data set used is based on images of a very large eye model and not pictures of actual eyes, thus results could vary when model is trained on a dataset of actual eyes.
2. Accuracy could be increased further if a CNN is used in tandem with the SVM, but that would require a larger dataset as well.
3. The model has not been tested on pictures of real people so the credibility of the study lessens due to that

2.1.1.3 Relationship to the proposed research work

This study is vital to our project as we intend to use the luminance based feature extraction method and SVM model proposed in this study to base our own model for detecting cataracts on. The difference being we will feed the model data gathered locally from hospitals and our classification will be more than just cataract or no cataract but also distinguish between types and stages of cataracts. But we want to base our approach on this study so that our end result is versatile, can be used in different environmental conditions and different smartphones and has a high accuracy.

2.1.2 Research Item 2

Chun et al., 2020

2.1.2.1 Summary of the research item

This study proposes a method for detecting refractive errors in pediatric patients using a deep learning network based detection method using smartphone images. In the study, photo refraction images were obtained of children using a smartphone camera with flash-light in dim lighting using a specific method to obtain a crescent like shape in the eye, over 300 images were obtained, and the patients were then tested by ophthalmologists using cycloplegic (dilated eyes using eye-drops) photorefractometry to obtain ground truth. Some of the dataset was augmented to increase the dataset quantity, after which 5 fold cross validation was used and the dataset was then used to train the ResNet-18 CNN model in order to classify between 7 ranges of refractive errors in the eye. Performance of different AI models was tested and the ResNet-18 model obtained the highest accuracy of 80 percent across all 7 ranges. The study found that with a limited dataset, an AI model could be trained to find ranges of refractive measurements in children which can be used to detect refractive errors with 80% accuracy, it also suggested that with a larger dataset, specific measurements of refractive errors could also be obtained.

2.1.2.2 Critical analysis of the research item

Strengths:

1. The research was done in real life environments and on people rather than models and was tested against results from ophthalmologist experts making the study very credible
2. The results covered a wide range of refractive errors.
3. Different models were tried and tested to obtain the best one

Weaknesses:

1. The methodology needed to obtain an image of the patient is tedious as it requires a room with little lighting (under 10LX) and for the image to be taken from a certain distance and angle to obtain the crescent shape in the eye using smartphone flashlight
2. Although the study was able to obtain a large range of refractive errors it could not make specific measurements as it lacked the dataset quantity to do so
3. The model was trained only on korean children and may not be able to perform well with other ethnicities.

2.1.2.3 Relationship to the proposed research work

This study is important to our project as it provides us with a tried and tested way of detecting refractive errors in children and also points us to the possible best model to do so as they have tested different models and shown their results, they have also provided images of heatmaps of patients eyes highlighting the most important areas for detection and could prove valuable when training our own model for detecting refractive errors. Lastly it provides us with a specific way of taking images to feed the model, namely by obtaining a crescent like shape in the eye using phone's flashlight.

2.1.3 Research Item 3

Sung et al., 2024

2.1.3.1 Summary of the research item

In this research study images of patients eyes were taken using two smartphone cameras in order to detect cataract in them using AI and were evaluated against the results from 3 ophthalmologists. The images were graded according to Emery Little (EL) classification which divides cataract into 3 stages, mild, moderate and severe. more than 3000 images were obtained from both smartphones (an Xperia and an Iphone) from a total of 178 participants, the study does not specify what ai model they used but it says they used the neural network libraries software to develop a cataract grading estimation model that classified the images into either mild/moderate or severe/very severe. The estimations were then compared against evaluations from 3 ophthalmologists on the patients using slit lamp microscopy which is the regular method for checking cataracts. They found that the they obtained an accuracy of 82% from both phones in classifying the images correctly

2.1.3.2 Critical analysis of the research item

Strengths:

1. The research was done on real patients and results were compared against ophthalmologists using the regular method for detecting cataracts which provides a useful insight into how the AI method fares against the regular method for cataract detection
2. The model is able to distinguish between mild/moderate and severe cataracts

3. The model was trained on images from different smartphones which makes it more versatile meaning it doesn't rely on the characteristics of a specific camera sensor

Weaknesses:

1. The study is not transparent about what model was trained
2. The study used more than 3000 images from only 178 participants making the chances of overfitting high
3. The model is only able to detect Nuclear sclerotic cataracts and the study does not mention whether it can detect the other two types of cataracts as well or distinguish between them

2.1.3.3 Relationship to the proposed research work

This study is essential to our project as it detects cataracts in a real world setting and compares against real results from ophthalmologists, allowing us to know the feasibility of the project especially under cases of patients having undilated eyes. As usual tests for cataracts involves patients eyes being dilated with eye drops, this study proves that an ai model does not need that in order to classify with good accuracy. Furthermore it tells us that classifying between different stages of cataracts is also possible.

2.1.4 Research Item 4**DO ET AL. (2022)****2.1.4.1 Summary of the research item**

In this study, a smartphone autorefraction technique was used to detect refractive errors in children's eyes and compared with the results from an autorefractor (an automated device to measure refractive errors). The smartphone based method does not use AI or machine learning, rather they take images from a smartphone from different angles and then calculate the refractive error, by calculating the pupil diameter size and the crescent image width reflected from the eye from the smartphone's flash. The study was conducted on 141 chinese schoolchildren, first the smartphone refraction technique was used and then it was compared to autorefraction technique. They compared the techniques under both Cycloplegic (drops given to eye for dilation) and non cycloplegic conditions. They found that the correlation between the two techniques was 88% for non-cycloplegic conditions and 80% for cycloplegic conditions, meaning they achieved impressive accuracy in detecting refractive measurements using smartphone images

2.1.4.2 Critical analysis of the research item

Strengths:

1. Only requires the use of smartphone and some calculations in order to measure refraction
2. High correlation to autorefractor(automated device) which is much more sophisticated.

Weaknesses:

1. The study does not mention what smartphone was used or more information about how they calculated the pupil diameter size and crescent width.
2. The study only mentions finding refractive errors for detecting Myopia, and does not talk about finding hypermetropia or astigmatism.
3. The study does not mention what conditions the images were taken under e.g lighting conditions etc.
4. Only tested on children, and since children can focus more easily with their eyes under non cycloplegic conditions it raises concerns whether this method is viable for adults.

2.1.4.3 Relationship to the proposed research work

This study is important as it outlines a method and some optical theory for finding refractive measurements without the use of ai, using only pupil diameter and crescent width from a smartphone image, using this method in tandem with ai could yield high accuracy and help us train the ai better. It also helps us in knowing how to take images to train the AI with as it is clear the crescent method for showing a crescent on the eye using smartphone flash is the best method for detecting refractive errors from smartphones, and that different angles of photos could be needed for higher accuracy.

2.1.5 Research Item 5

Qin et al. (2022-2023)

2.1.5.1 Summary of the research item

In this research a cross sectional study was conducted to assess whether an AI model can be used to check multiple eye diseases like myopia, strabismus, and ptosis early by analyzing photos taken by Smartphone cameras. A dataset of around 1400+ facial photographs from 476 childrens(47% females and 53% male) mostly ranging from age 6-12.A deep learning model was developed and its accuracy was estimated by using sensitivity, specificity, accuracy,(AUC),PPV,NPV,and N-LR and F1 score.GRADCAM++ was utilized to assess the impact of each region on model.

Before conducting the model training each subject underwent examination by ophthalmologist and were diagnosed by them.The study used CONVNext deep learning network for the detection of eye diseases.Before inputting the labelled images they were cropped into single eye local image using a square bounding box and for dual eye we used a rectangular bounding box for [myopia ptosis] and strabismus respectively. The network architect consist of 5 components, in which the first layer included the normalization layer and convulational layer followed by the three CNN blocks, later on the other components included the later three components of CNN blocks in which each begin their process by downsampling.

The model was good at detecting myopia with accuracy of 84%, strabismus of around 73%, and ptosis of around 85%. The model show similar performance accuracy among both sex groups of childrens.

2.1.5.2 Critical analysis of the research item

Strengths:

- First to use innovative approach to use face photograph of children and adolescent for the prediction of these three eye diseases simultaneously.
- Better accuracy was acheived with strong sensitivity in detecting myopian ptosis and strabismus.
- The dataset was that was used was large and and was gotten on spot, and was assesses by ophthalmologist, before performing the dataset collection.
- The dataset was quite diverse and included wide range of age groups and both genders.

Weaknesses:

- The local dataset that was gathered had aa limitation as the targeted audience was children's.(thus images used were gathered from under 18 subjects)

- The ideal scenario might cause ambiguity in real world scenario as the lightening will different in each environment.and camera positioning might be different in real world scenario.
- The data set collected was part of cross sectional study and it would be ambiguous as this does not state the condition of the study over some period of time.

2.1.5.3 Relationship to the proposed research work

This study might be useful in our project as it uses smartphone for the detection of myopia and other related eye diseases.The difference is we are not only using images gathered locally at one point of time but also using dataset gathered previously that is available online.our approach of dataset gathering locally is we are doing it continuously every week.

2.1.6 Research Item 6

Yuta et al 2024

2.1.6.1 Summary of the research item

This study aimed to develop an artificial intelligence algorithm capable of diagnosing cataracts and other various corneal diseases (a total of 36) using smartphone images. A total of 6,442 images were analyzed, with 6,106 captured via slit-lamp microscopy and 336 using smartphones. The AI model was trained on the slit-lamp images and validated against the smartphone images.

The AI algorithm managed to achieve an area under the ROC curve of 0.998 for normal eyes and high AUC values across various conditions, including 0.986 for infectious keratitis and 0.992 for cataracts.

The sensitivity and specificity for urgent referrals were both 1.00, indicating perfect accuracy, while performance for semi-urgent and routine categories also showed strong results.

2.1.6.2 Critical analysis of the research item

Strengths:

1. The AI demonstrated exceptional performance in identifying various corneal diseases and cataracts.
2. The use of smartphone images allows for broader accessibility in diverse settings.

Weaknesses:

1. The study primarily relied on slit-lamp images for training, which limits the representation of all possible imaging conditions encountered in real-world scenarios.
2. The AI performed well on the tested dataset, but its effectiveness remains to be validated across different populations and smartphone models.

2.1.6.3 Relationship to the proposed research work

The high performance metrics can serve as a benchmark for evaluating our app's algorithm. We learned that a diverse dataset for training and validation purposes is required to ensure that our app performs well across different demographics and conditions.

2.2 Analysis Summary of research items

Figure 2.1: Figure 1

Tabular Analysis

Sr.#	AUTHORS	DOI/REF NO	YEAR	JOURNAL	KEY FINDINGS	ACC %	LIMITATIONS
01	Askarian et al.	10.1109/JTEHM.2021.3074597.	2021	IEEE	- Algorithm Used: SVM - Trained on custom eye model - uses novel luminance based method - versatile and works for many environments and smartphones	96.6%	Does not detect type or stage of cataract
02	Yuta ueno et al	10.1136/bjoo-2023-324488	2024	BJO	- Algorithm used: YOLO V.5 - was able to differentiate between 36 eye diseases in 9 categories - was trained on slit lamp images and tested on smartphone images	75 %	- Device Limitation: Only works with iPhone 13 Pro. - Accuracy: Achieves 75% accuracy when predicting cataracts using smartphone images - Doesent detect Type of cataract
03	Jaemyoung Sung et al	10.1167/jovs.24-3715	2024	IOVS	- AI Model: Used Neural Network Librariesto make cataract grading model - Trained on slit lamp images and smartphone iamges from 2 smartphones - Uses images obtained from Xperia and Iphone to detect cataract and compared with ophthalmologic assessment	~84 %	- Not tested for different environmental conditions - Only detects moderate to severe cataracts - Doesent detect type of cataract

Figure 2.2: Figure 2

Sr.#	AUTHORS	DOI/REF NO	YEAR	JOURNAL	KEY FINDINGS	ACC %	LIMITATIONS
05	Chun et al	10.2196/16225	2020	<i>JMIR Med Inform</i>	• Detects Refractive errors using smartphone images accurately in children • trained on 305 photorefraction images • Uses Resnet-18 CNN • Categorized into seven refractive error classes.	81.6 %	• <u>Accuracy less than 90 percent</u> • <u>Requires dim lighting</u> • <u>Not tested on different smartphones</u>
06	Shu et al.	10.1001/jamanetworkopen.2024.25124	2024	<i>JAMA Netw Open</i>	• Detects pediatric diseases such as ptosis, myopia, and strabismus using smartphone images • trained on 1419 facial photographs of children • Uses the ConvNeXt38 CNN model • Evaluated model across different age and sex subgroups	~81 %	• <u>Does not detect or differentiate between different refractive errors such as hypermetropia and astigmatism</u> • <u>Not trained on different smartphones and environments</u>
07	Do et al.	Vol.63, 2819 – A0149.	2022	IOVS	• Tested detecting refractive errors in children using smartphone camera under cyclopegic and non cyclopegic conditions • Trained on images of 141 School Children aged 7 to 14 • Captured images in Various orientations which improved accuracy • Demonstrated strong correlation with autorefraction • Uses CNN	~77 %	• <u>Not tested on different smartphones under different conditions</u> • <u>Doesn't specify whether it differentiates between different refractive errors</u> • <u>Low accuracy</u>

Chapter 3

Proposed Approach

3.0.1 AI Model Approach

For our project we are making a mobile application that will have the capability to detect Types and stages of cataracts and refractive errors using images taken from the smartphone camera. Here i will present the methodology and framework we intend to use to accomplish the two main tasks.

1. Detecting Cataracts:

For this task we are using the following methodology taken from Askarian Et Al [1]

1. Image Acquisition:

Taking images of patients eye using the smartphone camera with flash on, with a distance of 10 to 50 cm from the eye, using the phones autofocus and cameras maximum resolution.

2. Preprocessing:

Preprocessing involves 2 main steps

- a. Image segmentation: With image segmentation we crop the image using a circle mask by marking the center of the circle and the radius of the circle to extract the region of interest this being the cornea of the eye,
- b. Denoising For denoising first we use median filter on the image to remove unnecessary glare and noise from the image. Secondly we apply watershed transformation to the image in order to extract the lens boundaries and to extract the ROI from the filtered image. Because most cataract diseases appear in the shape of a circle in the lens area, the watershed algorithm can be applied for image segmentation and visualization of the diseased areas.

3. Feature Extraction

Next we need to convert the image rgb values into luminance and brightness values since we are trying to detect cataracts from the luminance values reflected from the eye. In a

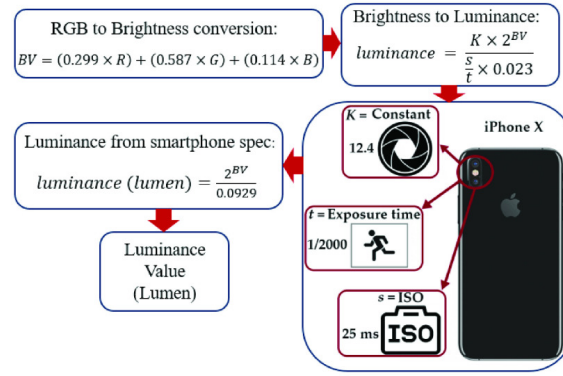


Figure 3.1: Enter Caption

smartphone, the luminance value can be determined by exposure time (), ISO speed (), light meter constant (), and brightness values (BV) these values are determined by each smartphones own specifications. we calculate brightness values BV of the image from its original rgb values using an algorithm. Next, by using the smartphone camera's specifications, the proposed algorithm converts the brightness into luminance values. The specific formulas and algorithms are shown in the following figure, specifically for IPHONE X. [1]

4. Classification

Next we use the luminance values for classification and differentiation between types and stages of cataracts. we are collecting our dataset locally from PIMS hospital OPD, after taking images of the patients using smartphone we obtain ground truth from ophthalmologists conducting slitlamp test on the patients to ascertain the type and stage of cataract. we then plan to train a support vector machine algorithm from this dataset, we might also augment some of the dataset to increase the quantity and use k fold cross validation. we also plan to test multiple AI models, particularly to try and use a cnn in tandem with the svm to see if we can increase the accuracy.

[1]

2. Detecting refractive errors:

For this task we are using the following methodology:

1. Image Acquisition:

For this step we have decided to try two ways of acquiring images and based on which method of image acquisition provides better accuracy in the model we will move forward with that. The first method is to take images of the eye from a certain angle in dim lighting using the smartphone camera's flash light such that a characteristic crescent like shape forms on the eye from the light reflected. This is taken from optical theory where in autorefractors use this method in order to detect refractive errors. The second method is to simply take pictures from a distance of 10 to 50 cm from the eye using smartphones

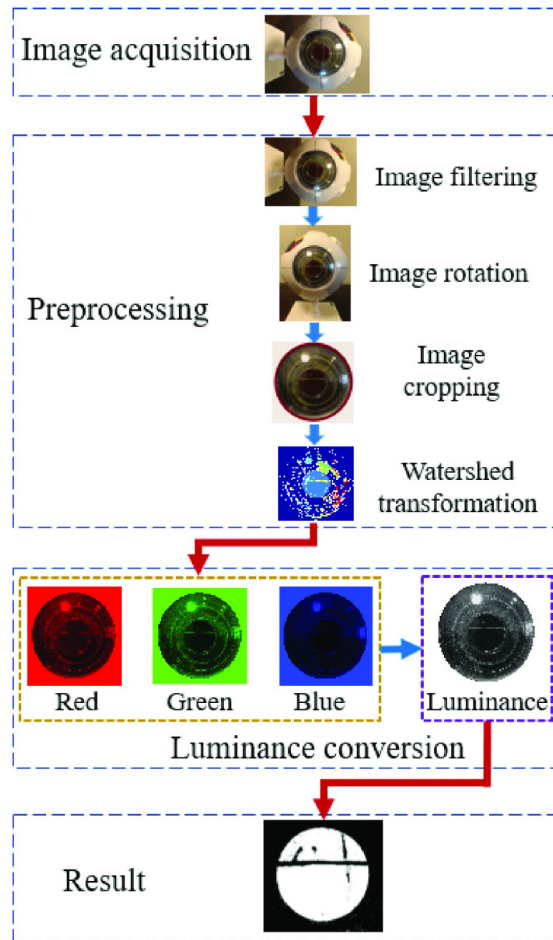


Figure 3.2: Enter Caption

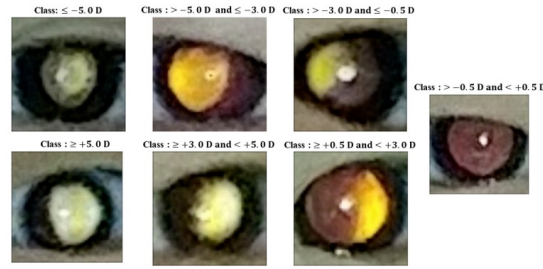


Figure 3.3: Enter Caption

flashlight under normal environmental conditions, this makes it easier for the end user but it might reduce the accuracy. Here are a few examples of images taken using the first method with the crescent shape. [4] 2. *Preprocessing*

Same as for detecting cataracts

3. *Feature Extraction:*

There has been no research to suggest that luminance values could prove beneficial and provide higher accuracy in detecting cataracts but given that the luminance method proposed by Askarian et al in their study is novel and recent, and since it highlights information reflected off the eye we have reason to suggest it might prove to improve the accuracy of the model, but it is as yet experimental. Thus we will try to use the luminance values from the images of refractive errors as well to see what effect it has on the model and whether it can improve its accuracy.

4. *Classification*

Next we use the luminance values, or the regular preprocessed images for classification of images, the end result we hope to achieve is to be able to get a precise measurement of the refractive error on the eye through which we can ascertain whether the patient has myopia, hypermetropia, or astigmatism, but as seen in other research papers it requires a large dataset to train the model to reach that point, other wise we intend to classify between ranges of refractive errors that will still give us the end result we want as our purpose is to detect myopia, hypermetropia and astigmatism and they do not require precise measurement to identify. we are collecting our dataset locally from Tanti Optics Optometry Shop, after taking images of the patients using smartphone we obtain ground truth from the optometrist conducting eye test using autorefractor on the patients to ascertain refractive error measurement. we then plan to train a RESNET-18 CNN model from this dataset, as it provided the highest accuracy in detecting refractive errors VS other ai models in a research study done by Chun et al. we might also augment some of the dataset to increase the quantity and use k fold cross validation. Heres the AI architecture we plan to use, taken from Chun Et Als study. [4] The photorefraction image inputs pass through 17 convolutional layers and one fully connected layer, and the outputs of the network assign the probabilities for each refractive error class given the image

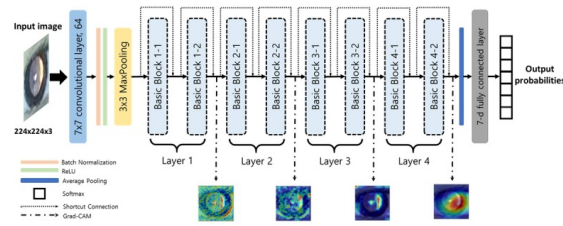


Figure 3.4: AI architect

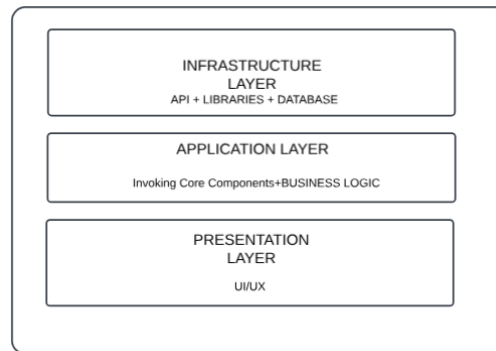


Figure 3.5: CLEAN ARCHITECT

3.0.2 Application Architect

In this Application we are developing using REACT NATIVE the Architect that is being followed is Clean Architect, That has three main components

- Infrastructure layer
- Application Layer(Business layer)
- Presentation Layer

Chapter 4

Conclusions and Future Work

4.0.1 Conclusion

In conclusion, our proposed product NAZAR app can provide an affordable, rapid, easy-to-use, and versatile method for detecting cataracts and other refractive error in eye's like myopia, astigmatism and hypermyopia by using smartphones without the use of bulky and expensive imaging devices. This method can easily be used and implemented for bedside telemedicine applications or in remote areas that have medical shortages and improve the health and well-being of people all over Pakistan.

4.0.2 Future Work:

The future work can be is the further integration and better accuracy of our models we are using in our application.

- **Accuracy:** To further train our model from the labelled data we are gathering from screening from our apps.
- **User Experience:** From the paractical input of our user we can enhance our application user flow and experience in our app.
- **Integration:** If our user desired we can add other feature and more eye diseases detection in our apps.

Chapter 5

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